COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/6/2026	**DATE SUBMITTED:**	10/07/2026
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ACTIVITY NO. 11.1 Containerization

1. PROCEDURE OUTPUT

1. Create a new repository for this activity.

```
keliste@Keliste-Workstation:~/CPE212_KELISTE$ cd HOA11.1_Containerization
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization$ ls
HOA11.1  README.md
```

2. Install Docker and enable the docker socket.

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ sudo apt remove $(dpkg --
get-selections docker.io docker-compose docker-compose-v2 docker-doc docker-buildx podman-docker contain
erd runc | cut -f1)
dpkg: no packages found matching docker.io
dpkg: no packages found matching docker-compose
dpkg: no packages found matching docker-compose-v2
dpkg: no packages found matching docker-doc
dpkg: no packages found matching docker-buildx
dpkg: no packages found matching podman-docker
dpkg: no packages found matching containerd
dpkg: no packages found matching runc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 36 not upgraded.
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$
```

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ sudo apt install ca-certi
ficates curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20260601~22.04.1).
ca-certificates set to manually installed.
curl is already the newest version (7.81.0-1ubuntu1.29).
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 36 not upgraded.
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$
```

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ sudo curl -fsSL https://d
ownload.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
```

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ sudo tee /etc/apt/sources
.list.d/docker.sources <<EOF
> Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Architectures: $(dpkg --print-architecture)
Signed-By: /etc/apt/keyrings/docker.asc
EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: jammy
Components: stable
Architectures: amd64
Signed-By: /etc/apt/keyrings/docker.asc
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$
```

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ sudo apt update
Get:1 https://download.docker.com/linux/ubuntu jammy InRelease [48.5 kB]
Get:2 https://download.docker.com/linux/ubuntu jammy/stable amd64 Packages [88.5 kB]
Hit:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Hit:5 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:6 http://ph.archive.ubuntu.com/ubuntu jammy-backports InRelease
Fetched 137 kB in 3s (53.6 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
36 packages can be upgraded. Run 'apt list --upgradable' to see them.
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$
```

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ sudo apt install docker-c
e docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  docker-ce-rootless-extras pigz
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras
  docker-compose-plugin pigz
0 upgraded, 7 newly installed, 0 to remove and 36 not upgraded.
Need to get 101 MB of archives.
After this operation, 383 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 https://download.docker.com/linux/ubuntu jammy/stable amd64 containerd.io amd64 2.3.6-1~ubuntu.22.
04~jammy [23.2 MB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy/universe amd64 pigz amd64 2.6-1 [63.6 kB]
Get:3 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce-cli amd64 5:29.8.2-1~ubuntu.
22.04~jammy [17.6 MB]
Get:4 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce amd64 5:29.8.2-1~ubuntu.22.0
4~jammy [24.4 MB]
```

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ sudo systemctl status doc
ker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2026-10-06 18:53:00 +08; 1min 49s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 5997 (dockerd)
      Tasks: 10
     Memory: 43.7M
        CPU: 2.351s
     CGroup: /system.slice/docker.service
             └─5997 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Oct 06 18:52:57 Keliste-Workstation dockerd[5997]: time="2026-10-06T18:52:57.551325048+08:00" level=inf>
Oct 06 18:52:57 Keliste-Workstation dockerd[5997]: time="2026-10-06T18:52:57.644852029+08:00" level=inf>
Oct 06 18:52:57 Keliste-Workstation dockerd[5997]: time="2026-10-06T18:52:57.666588274+08:00" level=inf>
Oct 06 18:52:59 Keliste-Workstation dockerd[5997]: time="2026-10-06T18:52:59.734792205+08:00" level=inf>
Oct 06 18:52:59 Keliste-Workstation dockerd[5997]: time="2026-10-06T18:52:59.814671557+08:00" level=inf>
Oct 06 18:52:59 Keliste-Workstation dockerd[5997]: time="2026-10-06T18:52:59.814893710+08:00" level=inf>
Oct 06 18:53:00 Keliste-Workstation dockerd[5997]: time="2026-10-06T18:53:00.566289237+08:00" level=inf>
Oct 06 18:53:00 Keliste-Workstation dockerd[5997]: time="2026-10-06T18:53:00.586947233+08:00" level=inf>
Oct 06 18:53:00 Keliste-Workstation dockerd[5997]: time="2026-10-06T18:53:00.587396051+08:00" level=inf>
Oct 06 18:53:00 Keliste-Workstation systemd[1]: Started Docker Application Container Engine.
```

3. Add to Docker group to your current user.

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ sudo usermod -aG docker $
USER
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ newgrp docker
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$
```

4. Create a Dockerfile to install web and DB server.

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ ls
dockerfile  start.sh
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ cat dockerfile
FROM ubuntu:22.04
ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && \
    apt-get install -y apache2 mariadb-server && \
    rm -rf /var/lib/apt/lists/*

COPY start.sh /start.sh
RUN chmod +x /start.sh

EXPOSE 80 3306
CMD ["/start.sh"]
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ cat start.sh
#!/bin/bash
service mariadb start
apachectl -D FOREGROUND
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$
```

5. Install and build the Dockerfile using Ansible.

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ cat docker_build.yml
---
- hosts: localhost
  become: true
  tasks:
    - name: Install Docker and Python Docker SDK
      apt:
        name:
          - docker-ce
          - docker-ce-cli
          - containerd.io
          - docker-buildx-plugin
          - docker-compose-plugin
          - python3-docker
        state: present
        update_cache: yes

    - name: Enable and start the Docker socket and service
      systemd:
        name: "{{ item }}"
        enabled: yes
        state: started
      loop:
        - docker.socket
        - docker.service

    - name: Add current user to docker group
      user:
        name: "{{ ansible_env.SUDO_USER | default(ansible_user_id) }}"
        groups: docker
        append: yes

    - name: Build the web and DB server image from the Dockerfile
      command: docker build -t web_db_server:latest {{ playbook_dir }}
      args:
        chdir: "{{ playbook_dir }}"
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$
```

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ ansible-playbook --ask-be
come-pass docker_build.yml
BECOME password:

PLAY [localhost] ***************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [localhost]

TASK [Install Docker and Python Docker SDK] ************************************************************
ok: [localhost]

TASK [Enable and start the Docker socket and service] **************************************************
ok: [localhost] => (item=docker.socket)
ok: [localhost] => (item=docker.service)

TASK [Add current user to docker group] ****************************************************************
ok: [localhost]

TASK [Build the web and DB server image from the Dockerfile] *******************************************
changed: [localhost]

PLAY RECAP *********************************************************************************************
localhost                  : ok=5    changed=1    unreachable=0    failed=0    skipped=0    rescued=0
 ignored=0

keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$
```

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization/HOA11.1$ docker images

                                                                                i Info →  U  In Use
IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest     5e2309035332       25.9kB         9.49kB    U
web_db_server:latest   54855ce448b2        616MB          116MB
```

6. Add, commit and push it to your repository.

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization$ git add HOA11.1
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization$ git commit -m "HOA11.1"
[main 20be9b7] HOA11.1
 3 files changed, 50 insertions(+)
 create mode 100644 HOA11.1_Containerization/HOA11.1/docker_build.yml
 create mode 100644 HOA11.1_Containerization/HOA11.1/dockerfile
 create mode 100644 HOA11.1_Containerization/HOA11.1/start.sh
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization$ git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.09 KiB | 1.09 MiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:AiconKeliste/CPE212_KELISTE.git
   6b789b7..20be9b7  main -> main
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA11.1_Containerization$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. What are the benefits of implementing containerizations?
   - Containerization packages an app with everything it needs, so it runs the same on any machine. Also, containers are lighter and faster to start than full VMs, so they save resources.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Claude AI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Checking for some commands**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used Claude Ai to check the commands that I used and I ask for some commands.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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